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IOP Publishing Journal of Optics
J. Opt. **23** (2021) 114001 (6pp) https://doi.org/10.1088/2040-8986/ac26ce

Helicity inversion and generation of orthogonal, degenerate index states of generic C points

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Received 27 July 2021, revised 3 September 2021
Accepted for publication 15 September 2021
Published 18 October 2021

Abstract
We discuss a technique to invert the helicity of generic C points and generate its orthogonal state of polarization in a single step through a fork grating (FG). It is experimentally demonstrated that the beam in each of its diffraction orders possesses a degenerate C point index state. The diffraction phenomenon reported is helicity-dependent, in which left and right-handed ellipse field singularities are separated. The orthogonal polarization state of the input singularity is produced in one of the first diffracted orders of the grating. For every singularity that the grating produces, its orthogonal polarization state is also available in one of its diffraction orders. In contrast to a half-wave plate, where helicity inversion is accompanied by index inversion, a FG inverts helicity without inverting the index. Therefore, a FG can be used to perform multi-functions under generic C point illumination.

Keywords: diffraction, polarization singularity, orbital angular momentum, orthogonal states, helicity

(Some figures may appear in colour only in the online journal)

1. Introduction

A beam with polarization singularity can be generated by the superposition of two circularly polarized orthogonal beams, each containing a vortex of different topological charges [1, 2]. In such a superposition, the phase difference between the beams is also helical, and together with the amplitude difference, the resulting beam is inhomogeneously polarized [3]. This means that the beam has spatially varying polarization distribution. The state of polarization (SOP) of the beam is circular at a particular point. This point, termed as a C point polarization singularity, is surrounded by polarization ellipses whose major axes rotate as one goes around the circular polarization point [2, 4]. They are characterized by a C point index (I_C) given by, $I_C = \frac{1}{2\pi}\oint \nabla\gamma \cdot dl$, where γ is the azimuth of the polarization ellipse. These beams are also known as Poincaré beams/spin-orbit beams [5–7].

These beams have drawn considerable interest in recent years due to their potential applications in various fields like optical trapping [8], beam manipulation [9], communication [10] and chirality measurement [11]. They have also been recently used to substantiate the geometric spin Hall effect of light [12, 13]. The optical helicity of these beams acts as a promising observable along with the angular momentum

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content of the beam to gain an in-depth understanding of light-matter interactions based on symmetries [14, 15]. Orthogonal states of polarization singularities and basis construction with them have been the topic of the study recently [16]. Due to the highly rich polarization inhomogeneity of a Poincaré beam, deriving an orthogonal state and simultaneously producing its degenerate index states is a non-trivial problem. In the present study, we introduce a method of generating an orthogonal state of a given generic C point and also producing a set of its degenerate index states in a single step by employing a fork grating (FG) as a key element.

A normal grating acts as a simple beam splitter with multiple outputs, and each beam has the same polarization distribution as that of the input [17]. Further, the field distribution in each diffraction order is also the same, but with the increase in diffraction order, the amplitude goes on decreasing. A FG, on the contrary, when illuminated by a plane wave, produces a vortex beam in each of its orders depending on the charge of the FG [18–20]. FG can generate phase singular beams, each possessing a different topological charge in its diffraction orders. However, there is no change in the polarization distribution of the homogeneously polarized scalar wave. Meanwhile, multiple attributes of a polarization singularity undergo changes on diffraction through a FG. Interestingly, some of the topological parameters are also preserved. In this paper, we list and discuss hitherto unexplored, counter-intuitive, multiple properties exhibited by a FG under generic Poincaré beam illumination. These properties are useful to realize helicity inversion and generation of orthogonal, degenerate index states of generic C points.

In our study, we use a binary amplitude FG to diffract C point. The diffraction process is shown to be helicity-dependent, where C points with opposite handedness are separated. For every singularity that the grating produces, its orthogonal polarization state is also available in one of its diffraction orders.

2. Helicity and orthogonal states of spin-orbit beams

Consider a spin-orbit beam represented as a superposition state given by

$$\overrightarrow{E}=\frac{1}{\sqrt{2}}\left\{(\hat{x}+i\hat{y})(x+iy)^{m}+(\hat{x}-i\hat{y})(x+iy)^{n}\exp(i\theta_0)\right\}, \tag{1}$$

where m and n are the topological charges of the superposing phase singular beams, and $\frac{1}{\sqrt{2}}(\hat{x}+i\hat{y})$ and $\frac{1}{\sqrt{2}}(\hat{x}-i\hat{y})$ are right and left circular basis vectors. θ_0 is the initial phase factor. This beam has a C point at its center when $m\neq n$. For the field $\overrightarrow{E}$ given by equation (1), the C point index is $I_C=(n-m)/2$. Dark C points are generated when $m\neq n$ and both m and n are non-zero. For bright C points, either m or n is zero. The generic bright C points are classified as lemon, star, and monstar based on their polarization distributions [4, 21]. These polarization singularities are generally identified by constructing the Stokes field $S_{12}=S_1+iS_2$. This is a mathematically constructed complex field using the Stokes parameters S_1 and S_2. The phase distribution of this complex field is given by, $\phi_{12}=\tan^{-1}\left(\frac{S_2}{S_1}\right)$, which equals twice the azimuth (γ) value [7]. The C point index I_C is related to Stokes index $\sigma_{12}=\dfrac{\Delta\phi_{12}}{2\pi}=2I_C$, where $\Delta\phi_{12}$ is the accumulated ϕ_{12} phase around the polarization singularity. Therefore, phase vortices of S_{12} Stokes field are polarization singularities.

For an inhomogeneously polarized field, all the four Stokes parameters are spatially varying.

Let us now discuss the various attributes of C points in the incident and diffracted beams. Along with spin (**S**), helicity (h) or handedness of a photon is an important parameter that can be defined [22, 23] by the sign of the spin projection on the momentum vector (**P**) as,

$$h=\frac{\mathbf{S}\cdot\mathbf{P}}{|\mathbf{S}||\mathbf{P}|}. \tag{2}$$

For a photon, helicity takes value $+1$ and -1 for right and left handedness respectively. A left elliptically polarized light, under circular basis decomposition, has a dominant left circular polarization (LCP) component (spin component) in comparison with its right circular polarization (RCP) component. For a polarization singularity located at the core of the beam, helicity is decided by the handedness of the component vortex beam, which possesses a lower orbital angular momentum (OAM) value in the superposition. This is because the vortex with a lower charge has dominant amplitude near the core, in comparison with a vortex with a higher charge. For example, a left-handed (LH) lemon can be generated due to the superposition of a non-vortex beam and a positively charged scalar vortex in LCP and RCP states, respectively. Therefore, helicity can be controlled by OAM transfer processes in spin-orbit beams [16, 24].

Two polarization distributions are orthogonal if their inner product is zero, i.e. $\langle\psi|\psi^o\rangle=0$, where ψ and ψ^o are orthogonal polarization states. Apart from that, there is point-wise orthogonality existing between the two distributions. For homogeneously polarized beams, both helicity inversion and orthogonal state conversion can be achieved using a half-wave plate (HWP) [25]. On the contrary, for polarization singular beams, the use of the HWP results in both helicity and index inversion [26]. The conversion to an orthogonal state for these beams involves helicity inversion and rotation but not index inversion [16], and hence, the use of a HWP alone will not suffice. The spatially varying Stokes parameters $S_1(x,y), S_2(x,y)$, and $S_3(x,y)$ invert their sign simultaneously for orthogonal pair realization. Here, we are dealing with the polarization degrees of freedom alone. Figure 1 shows the orthogonal polarization pairs of spin-orbit beams. The right-handed (RH) and LH states of polarization are depicted in red and blue colors, respectively, in the entire paper. The orthogonal state for a C point singularity has been realized using a spiral phase plate and a Pancharatnam phase shifter (a combination of two HWPs) [16]. In this paper, we show that the polarization singularities in orthogonal pairs can be generated in a single step using a FG.

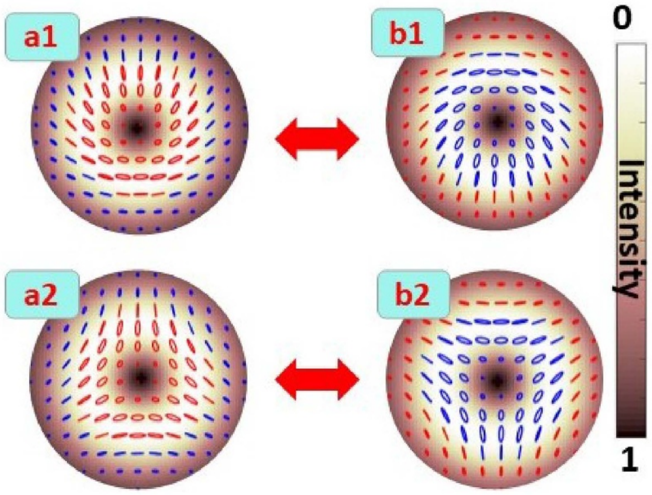

Figure 1. Orthogonal pairs of C points. (a1), (b1) and (a2), (b2) are orthogonal pairs of lemons and stars respectively. The right-handed and left-handed states of polarization are depicted in red and blue colors respectively.

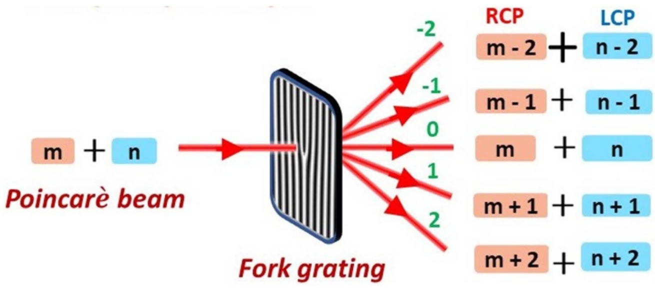

Figure 2. Schematic representation of diffraction of a Poincaré beam through a binary amplitude FG of charge 1.

3. FG illuminated by a polarization singularity

The behavior of FG is highly dependent on the nature of illumination. Under C point beam illumination, the FG exhibits its hitherto unexplored, counter-intuitive, multiple properties. Before dealing with this case, first we revisit the use of FG in scalar optics. Among the various methods of vortex generation [27, 28], the use of FG is the most common one [29, 30]. A FG can be fabricated holographically by recording the interference of a plane wave and a helical wave with a tilt between them. Such a holo-FG, with a sinusoidal amplitude profile, produces $+1$, 0, -1 diffraction orders. To get multiple diffraction orders, FGs with different transmittance (sinusoidal phase or binary phase/amplitude) profiles can be used.

Consider that the transmittance function t_g of the binary amplitude FG is given by,

$$t_g = A + B\,\mathrm{sgn}[\cos(Gx - l\theta(x,y))] \tag{3}$$

where $G = 2\pi/\Lambda$ is the grating constant, and Λ is the grating period. l is the topological charge, $\theta(x,y)$ is the azimuthal phase variation of the vortex in the first diffraction order, and A and B are constants. The function $\mathrm{sgn}(x)$ is a signum function defined as

$$\begin{aligned}\mathrm{sgn}(x) &= -1 \quad \text{for} \quad x \leqslant 0\\ &= +1 \quad \text{for} \quad x > 0.\end{aligned} \tag{4}$$

The binary amplitude grating given in equation (3) is capable of producing multiple diffraction orders. The Fourier series expansion for the grating is given by,

$$\Sigma_{p=-\infty}^{\infty} C_p \exp\{ip(Gx - l\theta)\} \tag{5}$$

where the integer p denotes the diffraction order number. C_p is the relative coefficient of the p^{th} diffraction order, which relates to the diffraction efficiency. As one can see, there is a vortex in each of the diffraction orders, and the charge of the vortex in each diffraction order is given by pl. This means in the first-order, one can get a vortex beam of charge l, and in second-order vortex beam of charge $2\,l$ and, so on. When illuminated by a scalar vortex beam with charge m, such a grating will produce a non-vortex beam with a bright spot at p^{th} order provided $pl - m = 0$. In other $\pm p^{\text{th}}$ orders, vortices of order $(m \pm pl)$ will be observed.

Since polarization singularities can be generated by the superposition of two scalar beams in orthogonal spin angular momentum (SAM) states with appropriate OAM contents, the action of a FG on them can be understood by the application of first principles. Using the superposition principle, the action of FG is analyzed. This means that the input light can be decomposed into its polarization components, and the effect of the FG on individual components can be combined together to get the SOP distribution at the diffraction order.

3.1. Helicity inversion

Polarization singular beams with the superposition of OAM states and SAM states as in equation (1), upon diffraction, will have the resultant superposition of optical vortices of charge $(m \pm pl)$ and $(n \pm pl)$ in orthogonal SAM states in the $\pm p^{\text{th}}$ diffraction order (figure 2). Simulated results for the diffraction of all the four different generic C points from a FG of charge one from -2 to $+2$ diffraction orders are shown in figure 3. For example, when the input beam is a RH star with $m = 0$ and $n = -1$, the output beam after diffraction from a FG with charge 1 ($l = 1$), generates polarization singular beams with OAM combination of $(-2,-3)$, $(-1,-2)$, $(0,-1)$, $(1,0)$ and $(2,1)$ from $p = -2$ to $p = 2$ orders respectively as shown in figure 3(a). Since the OAM combination in respective spin states remains the same, the zeroth-order has the same polarization distribution as that of the incident beam. The OAM content of the diffracted orders is dependent on the input C point index (I_C), the charge of the grating (l) and the diffraction order (p). As helicity is decided by the handedness of the component vortex beam, which possesses a lower OAM value in the superposition, the negative (positive) diffraction orders are RH (LH) for positive input C point index. For the case of negative input C point index, the helicity in the positive and negative diffracted orders reverse. The helicity in the positive and negative orders are opposite to each other and can also be inferred

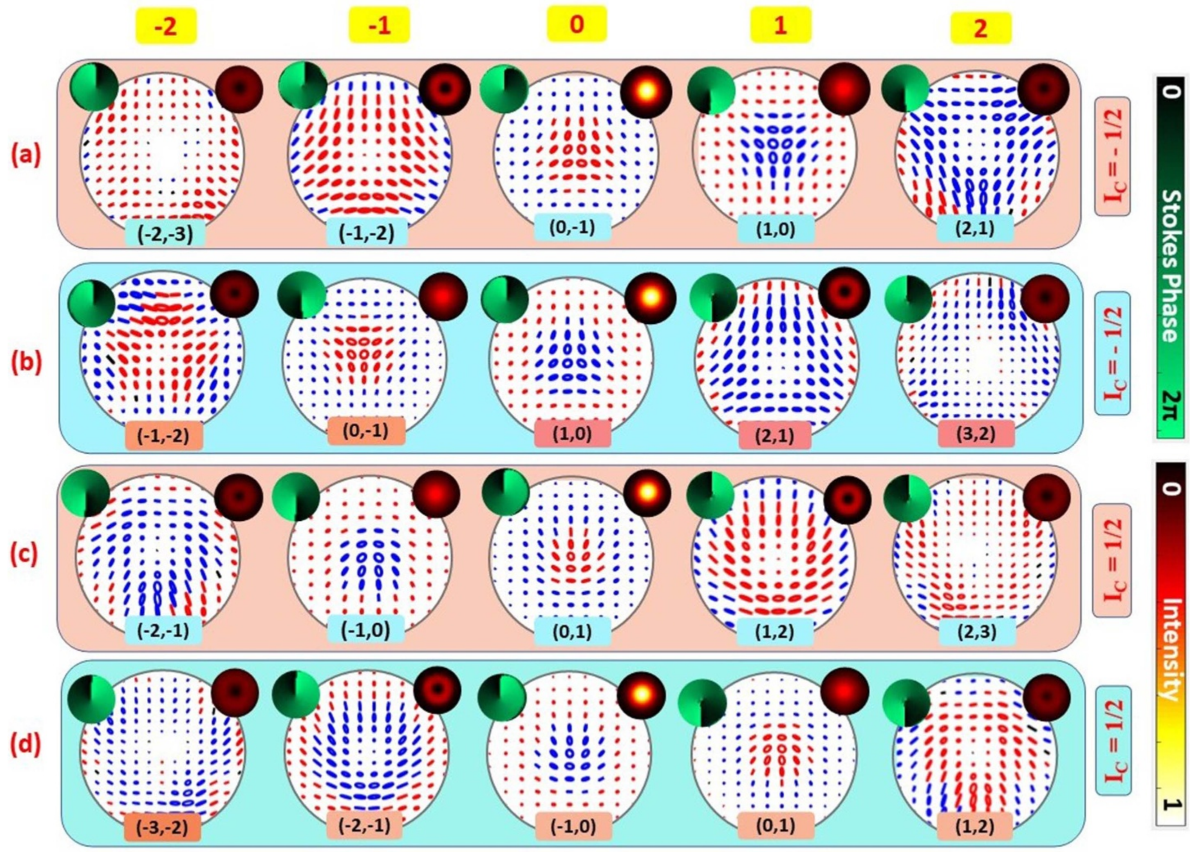

Figure 3. Simulated results showing index preserved helicity inversion in diffraction of C points through FG. Diffraction orders (-2 to $+2$) when incident C point is (a) RH star (b) LH star (c) RH lemon (d) LH lemon. SOP distribution and corresponding OAM content are shown. Insets show intensity pattern and Stokes phase around the C point.

from the Stokes parameter $S_3(x,y) = +1$ or -1 for RH and LH polarizations, respectively. The diffraction segregates the degenerate C points as per the sign of the helicity in positive and negative diffraction orders.

3.2. Orthogonal states generation

The orthogonal polarization state of the input generic C point is generated in one of the first diffraction orders. For every polarization singularity that the grating produces, its orthogonal polarization state is also available in one of its diffraction orders. For the generic C points generated from a negatively charged phase singular beam, i.e. $(-1,0)$ or $(0,-1)$ OAM combinations, it is observed that p^{th} order and $(-p+1)^{\text{th}}$ orders are orthogonal to each other. Similarly, for the generic C points generated from a positively charged phase singular beam, i.e. $(1,0)$ or $(0,1)$ combinations, it is observed that p^{th} order and $(-p-1)^{\text{th}}$ orders are orthogonal to each other.

The C points present in the positive and negative diffracted orders are π phase-shifted. This can be explained as the Gouy phase difference between the C points in the positive and negative orders. The Gouy phase for a beam with a vortex [31, 32] of charge m is given by $(|m|+1)\tan^{-1}(\frac{z}{z_R})$. Therefore, the Gouy phase for a polarization singularity [33] is given by $(|m|-|n|)\tan^{-1}(\frac{z}{z_R})$. In the far-field ($z \to \infty$), this value equals $(|m|-|n|)\pi/2$. Hence, for the diffraction of generic C point singularities, the Gouy phase for the C points in the positive and negative diffracted orders are $\pm\pi/2$ and $\mp\pi/2$, respectively. This gives the overall Gouy phase difference of π between the C points in the positive and negative diffracted orders.

As mentioned before, the orthogonal state generation requires helicity inversion followed by a rotation of the SOPs. So, this Gouy phase shift provides the required rotation, which is tantamount to Pancharatnam phase shift necessary for orthogonal state realization.

3.3. Degenerate index states generation

The transfer of OAM in the diffraction process results in C point—C point transformation, each with the same I_C index as that of the input C point, but different attributes such as handedness, bright or dark intensity due to different OAM compositions in respective SAM states. The diffraction process results in OAM transfer in component vortex beams of the C point. The transfer process is such that in a given diffraction order, the increase (decrease) in the OAM content for the RH (LH) vortex beam is accompanied by the decrease (increase) in the OAM content in the LH (RH) vortex beam so that the difference $(n-m)$ remains same. Hence, this results in degenerate index states generation.

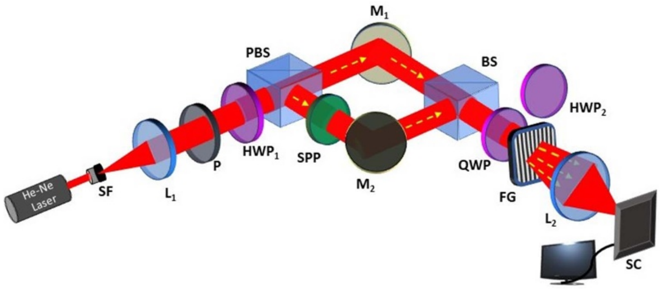

Figure 4. Experimental setup showing diffraction of C points from FG. SF: spatial filter, P: polarizer, BS: beam splitter, PBS: polarizing beam splitter, M1, M2: mirrors, L1, L2: lenses, QWP and HWP: quarter and half wave plate, SPP: spiral phase plate, SC: Stokes camera.

4. Experimental verification

In the experimental set-up shown in figure 4, a He–Ne laser source of wavelength 632.8 nm is used. The laser beam is spatially filtered, collimated, and passed through a polarizer (P) and HWP_1 to obtain 45° linearly polarized light. It is launched into a modified Mach–Zehnder set-up consisting of a spiral phase plate (*SPP*) in one of the arms to get desirable OAM superposition in orthogonal polarization states. A quarter wave plate (*QWP*) at 45° is introduced after the interferometer, to ensure superposition in the circular basis. For the *SPP* with charge 1, this leads to the generation of a RH lemon. We use flipping of SPP at its position, which changes the charge of the input phase singular beam from 1 to -1. Flipping of SPP, thus generates a RH star. By inserting a HWP (HWP_2), just after the QWP, we can obtain the other possible generic C point(s), the left handed star (lemon) from the RH lemon (star) through index sign inversion [26].

The generated beam is then passed through a FG of charge 1 to obtain polarization singular beams with different OAM combinations in different diffraction orders by the OAM transfer process. Polarization distributions of various diffracted orders are obtained through Stokes camera (Salsa Full Stokes camera 1040 × 1040 by Bossa Nova) placed at the focal plane of lens L_2. This camera uses a combination of a quarter wave plate and a polarizer at different pre-determined orientations to obtain intensity distributions in different component polarization states. All the captured frames are processed by the software of the camera and the corresponding Stokes parameters are obtained. The polarization pattern is obtained by plotting the polarization ellipses through MATLAB based on the obtained Stokes parameters. Corresponding Stokes phase ϕ_{12} (azimuth variation) is also obtained to verify the polarization distribution of the diffracted singular beams obtained experimentally. The experimentally observed polarization patterns are shown in figure 5. The orthogonal pair of the input beam is also observed in one of the first orders. The experimental results of the diffraction of all the generic C points shown in figure 5 are in reasonable agreement with the simulated results.

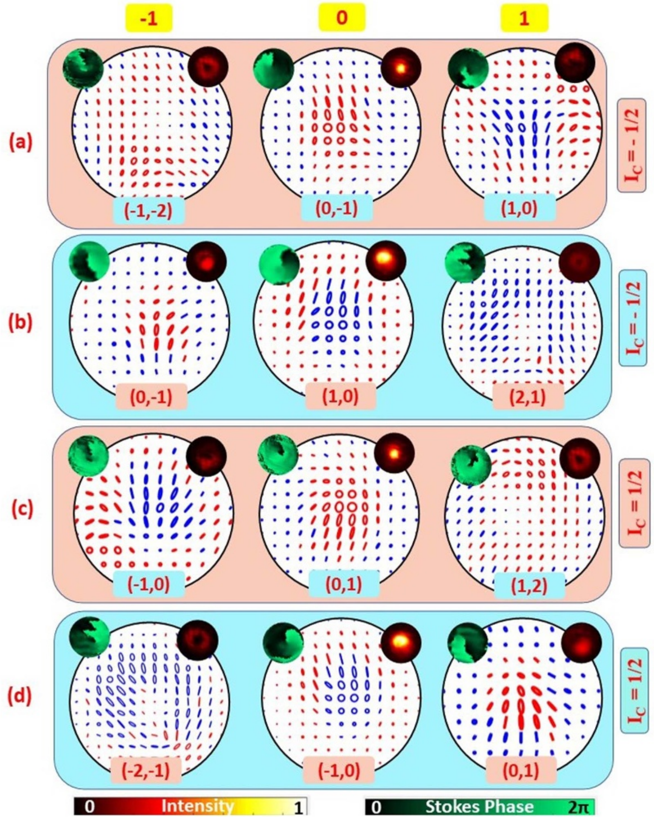

Figure 5. Experimental results showing index preserved helicity inversion through diffraction of C points ((a) RH star (b) LH star (c) RH lemon (d) LH lemon) from a FG. SOP distribution and the corresponding OAM content in the diffracted orders (-1 to $+1$) are shown. Insets show intensity pattern and Stokes phase around the C point.

5. Conclusion

In conclusion, we have demonstrated a simple diffraction technique to invert the helicity of generic C point beams. This inversion is achieved by the use of a binary amplitude FG. Concomitant interesting properties that we have discussed make this diffraction process very special. The orthogonal pair for the input beam is obtained in one of the first diffracted orders. Therefore, FG can be used as a single element to generate orthogonal polarization singular state. The C point index in all the diffracted orders is degenerate in the process. The C points in the positive and negative diffracted orders are observed to be π-phase shifted due to the Gouy phase. The transfer of OAM due to FG is such that C points in all the diffraction orders form orthogonal pairs of polarization singularities. These properties witnessed on diffraction of generic C points through a FG enable easy manipulation of these beams for their intended applications.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

Acknowledgment

S Deepa acknowledges Women Scientist Fellowship of Department of Science and Technology, Ministry of Science and Technology, India (SR/WOS-A/PM-66/2017).

Baby Komal acknowledges UGC-SRF fellowship.

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References

[1] Maurer C, Jesacher A, Fürhapter S, Bernet S and Ritsch-Marte M 2007 *New J. Phys.* **9** 78
[2] Freund I 2002 *Opt. Commun.* **201** 251–70
[3] Ruchi, Senthilkumaran P and Pal S K 2020 *Int. J. Opt.* **2020** 2812803
[4] Dennis M 2002 *Opt. Commun.* **213** 201–21
[5] Beckley A M, Brown T G and Alonso M A 2010 *Opt. Express* **18** 10777–85
[6] Galvez E J, Khadka S, Schubert W H and Nomoto S 2012 *Appl. Opt.* **51** 2925–34
[7] Arora G, Ruchi and Senthilkumaran P 2019 *Opt. Lett.* **44** 5638–41
[8] Zhan Q 2004 *Opt. Express* **12** 3377–82
[9] Shi P, Du L and Yuan X 2018 *Opt. Express* **26** 23449–59
[10] Milione G, Nguyen T A, Leach J, Nolan D A and Alfano R R 2015 *Opt. Lett.* **40** 4887–90
[11] Samlan C T, Suna R R, Naik D N and Viswanathan N K 2018 *Appl. Phys. Lett.* **112** 031101
[12] Ling X, Zhou X and Yi X 2017 *Opt. Commun.* **383** 412–7
[13] Bliokh K Y, Prajapati C, Samlan C T, Viswanathan N K and Nori F 2019 *Opt. Lett.* **44** 4781–4
[14] Tischler N, Fernandez-Corbaton I, Zambrana-Puyalto X, Minovich A, Vidal X, Juan M L and Molina-Terriza G 2014 *Light: Sci. Appl.* **3** e183
[15] Angelsky O V, Bekshaev A Y, Hanson S G, Zenkova C Y, Mokhun I I and Jun Z 2020 *Front. Phys.* **8** 114
[16] Ram B S B, Ruchi and Senthilkumaran P 2018 *Opt. Lett.* **43** 2157–60
[17] Born M and Wolf E 2019 *Principles of Optics: 60th Anniversary Edition* 7th edn (Cambridge: Cambridge University Press)
[18] Moreno I, Davis J A, Pascoguin B M L, Mitry M J and Cottrell D M 2009 *Opt. Lett.* **34** 2927–9
[19] Moreno I, Davis J A, Ruiz I and Cottrell D M 2010 *Opt. Express* **18** 7173–83
[20] Janicijevic L and Topuzoski S 2008 *J. Opt. Soc. Am.* A **25** 2659–69
[21] Berry M V and Hannay J H 1977 *J. Phys. A: Math. Gen.* **10** 1809–21
[22] Bliokh K Y and Nori F 2015 *Phys. Rep.* **592** 1–38
[23] Cameron R P, Barnett S M and Yao A M 2012 *New J. Phys.* **14** 053050
[24] Deepa S, Ram B S B and Senthilkumaran P 2019 *Sci. Rep.* **9** 12491
[25] Goldstein D H 2011 *Polarized Light* (Boca Raton, FL: CRC Press)
[26] Pal S K, Ruchi and Senthilkumaran P 2017 *Appl. Opt.* **56** 6181–90
[27] Beijersbergen M, Coerwinkel R, Kristensen M and Woerdman J 1994 *Opt. Commun.* **112** 321–7
[28] Heckenberg N R, McDuff R, Smith C P and White A G 1992 *Opt. Lett.* **17** 221–3
[29] Basistiy I, Bazhenov V, Soskin M and Vasnetsov M 1993 *Opt. Commun.* **103** 422–8
[30] Matta S, Vayalamkuzhi P, Naik D N and Viswanathan N K 2018 *J. Opt.* **20** 075604
[31] Kawase D, Miyamoto Y, Takeda M, Sasaki K and Takeuchi S 2008 *Phys. Rev. Lett.* **101** 050501
[32] Philip G M, Kumar V, Milione G and Viswanathan N K 2012 *Opt. Lett.* **37** 2667–9
[33] Cardano F, Karimi E, Marrucci L, de Lisio C and Santamato E 2013 *Opt. Express* **21** 8815–20